

BRAC UNIVERSITY
Merul Badda, Dhaka, Bangladesh
CSE331 : Automata and Computability

Quiz 1
Summer 2024

Duration: 30 minutes

CO1

Total: 20 marks

Name:	ID:	Section:
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$L_1 = \{w \in \Sigma^* \mid w \text{ starts with } 10\}, \text{ where } \Sigma = \{0, 1\}$
 $L_2 = \{w \in \Sigma^* \mid w \text{ contains an even number of } 0\text{s}\}, \text{ where } \Sigma = \{0, 1\}$

RE

1. Write the Regular Expressions for the following regular languages: (5 x 2 = 10 marks)

L(M) $\rightarrow$ $\{w \in \Sigma^* \mid w \text{ contains exactly three } 0\text{s}\}, \text{ where } \Sigma = \{0, 1\}$

Solution:

$1^* 0 1^* 0 1^* 0 1^*$

L(M) $\rightarrow L_1$

Solution:

$10\Sigma^*$

DFA

2. Draw the state diagram of the DFA for the following regular languages: (5 x 2 = 10 marks)

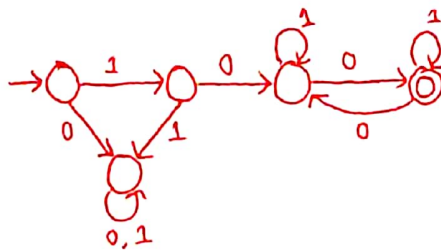
$L(M) \rightarrow L_2$

Solution:



$L(M) \rightarrow L_1 \cap L_2$

Solution:



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Quiz 2
Spring 2024

Duration: 30 minutes

CO2

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Name:

ID:

Section:

NFA

1. Draw the state diagram of the NFA for the following regular language: (5 marks)

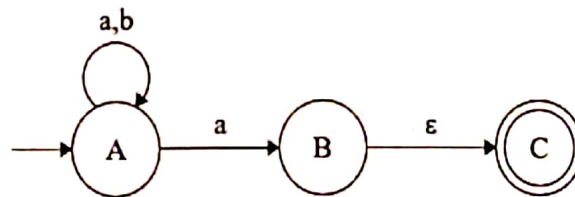
$L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ starts baaab or bab}\}$, where $\Sigma = \{a, b\}$. (Use only 6 states)

Solution:

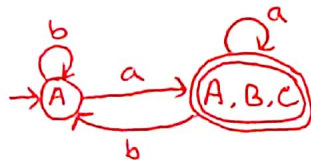


NFA $\rightarrow$ DFA

2. Convert the following NFA into equivalent DFA:



Solution:

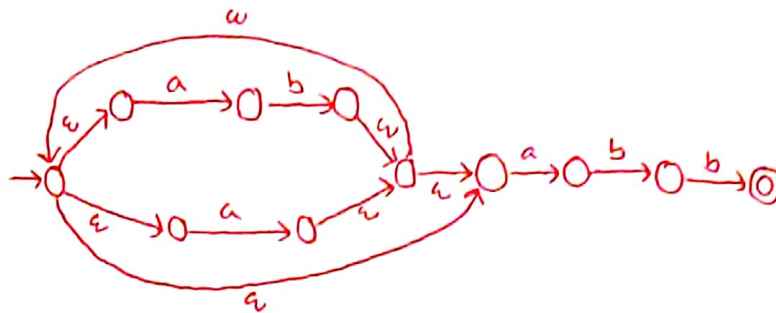


RE $\rightarrow$ NFA

3. Convert the following RE into equivalent NFA:

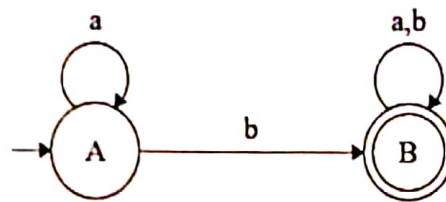
$L(M) \rightarrow (ab \mid a)^*abb$

Solution:



DFA $\rightarrow$ RE

4. Convert the following DFA into equivalent RE:



Solution:

$a^*b(a|b)^*$

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Quiz 3
Summer 2024

Duration: 25 minutes

CO3

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CFG

1. Convert the following RE into CFG: (10 x 1 = 10 marks)

$10^*(0|01)^*$

Solution:
 $S \rightarrow AB$
 $A \rightarrow YD$
 $D \rightarrow 0D \mid \epsilon$
 $B \rightarrow EB \mid \epsilon$
 $E \rightarrow X \mid XY$
 $X \rightarrow 0$
 $Y \rightarrow 1$

2. Remove ambiguity from the following CFG: (10 x 1 = 10 marks)

$$E \rightarrow E + E$$
$$E \rightarrow E * E$$
$$E \rightarrow a$$

Solution:
 $E \rightarrow E + T \mid T$
 $T \rightarrow T * F \mid F$
 $F \rightarrow a$

3. Construct CFG for the following CFL: (10 x 1 = 10 marks)

$L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains balanced parentheses.}\}$, where $\Sigma = \{ (,), \{, \}, [,] \}$.

Solution:

$S \rightarrow (S) \mid \{S\} \mid [S] \mid SS \mid \varepsilon$

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Quiz 4

Summer 2024

Duration: 30 minutes

CO4

Total: 30 marks

Name:	ID:	Section:
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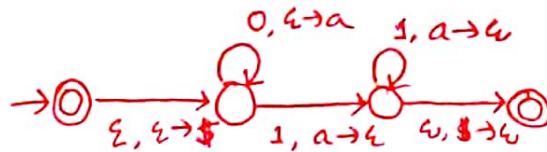
PDA

Let, $\Sigma = \{0, 1\}$.

1. Draw the state diagram of a PDA for the following CFL over Σ : (10 x 1 = 10 marks)

$L(M) \rightarrow \{w = 0^n 1^m \mid n, m \geq 0 \text{ and } n = m\}$.

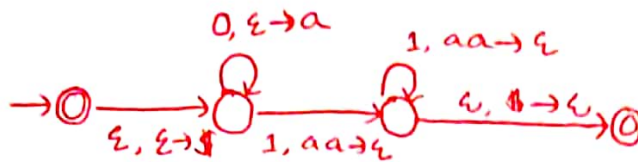
Solution:



2. Draw the state diagram of a PDA for the following CFL over Σ : (10 x 1 = 10 marks)

$L(M) \rightarrow \{w = 0^n 1^m \mid n, m \geq 0 \text{ and } n = 2m\}$.

Solution:



3. Draw the state diagram of a PDA for the following CFL over Σ : (10 x 1 = 10 marks)

$L(M) \rightarrow \{w = 0^n 1^m \mid n, m \geq 0 \text{ and } n > 2m\}$.

Solution:

